MIL-P-47226(MI)
12 July 1974
SUPERSEDING
MIS 15025
10 March 1965

MILITARY SPECIFICATION

PLATING, COPPER, ELECTROLESS (FOR NON-CONDUCTING MATERIAL)

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for chemically-deposited conductive copper films used to coat electrically non-conducting materials.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

L-T-90	Tape, Pressure Sensitive, Adhesive, (Cellophane and Cellulose Acetate)

Military

MIL-C-14550	Copper Plating (Electrodeposited)

(Copies of specifications, standards, publications, and drawings, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 95GP

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3. REQUIREMENTS

3.1 Material. The electroless copper film shall consist of a homogeneous coating of 99 percent copper produced by chemical reduction in a water solution of complexed copper ions.

3.2 Adhesion to substrate. The electroless copper film shall adhere, without lifting or scaling, to a suitably deglazed and chemically prepared substrate. (See 4.2.2 and 4.5.2)

3.3 Workmanship. The electroless copper film shall be free of blisters and voids. (See 4.5.1)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. The quality conformance inspection shall be conducted on each lot of plated parts offered for acceptance.

4.2.1 Inspection lot. An inspection lot shall consist of plated parts of the same type, plated under similar conditions, and submitted for inspection at one time.

4.2.2 Testing. Three test specimens shall be prepared with minimum dimensions 0.04 by 1 by 4 inches from a material equivalent to that of the parts represented. Equivalent material shall include chemical composition, condition, temperature treatment, and surface finish prior to plating. These specimens shall be introduced into the lot prior to the cleaning and other surface-preparation operations before plating and shall not be separated therefrom until after completion of processing. The specimens shall be coated with electroless copper film and electroplated with copper in accordance with MIL-C-14550, concurrently with the parts represented. These specimens shall be subjected to the tests in Table I.

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4.2.3 Action in case of failure. If a failure occurs, the lot represented by the specimens shall be rejected. The supplier shall take corrective action as necessary on all parts plated under essentially the same conditions and which are considered subject to the same type of failure. Acceptance of further lots may be withheld until three specimens which incorporate the corrective action have been subjected to and have passed the quality conformance inspection.

TABLE I

Quality conformance inspection		
Examination or test	Requirement paragraph	Method paragraph
Visual examination	3.3	4.5.1
Adhesion to substrate	3.2	4.5.2

4.3 Inspection equipment and facilities. Inspection equipment shall be of adequate quality and quantity to permit performance of the required examinations and tests, and shall comply with the equipment described in the applicable inspection methods of this specification and the documents referred to therein. Inspection facilities shall be adequate to permit performance of the required inspection.

4.4 Standard inspection conditions. Unless otherwise specified, all inspections shall be performed under conditions of room ambient pressure, temperature and humidity.

4.5 Inspection methods.

4.5.1 Visual examination for workmanship. The plating shall be examined under 10 magnification for compliance with the requirements specified in 3.3.

4.5.2 Test for adhesion to substrate. A sharp knife shall be used to cut a 1/2 by 3/4 inch rectangle through the plating on the test specimens. A strip of cellophane tape, 1/2 inch wide by 2 inches long, conforming to L-T-90, shall be pressed firmly across the rectangular cut on the surface of the copper plating. The tape shall be removed quickly by hand, approximately perpendicular to the surface. Compliance with the requirement specified in 3.2 shall be noted.

5. PREPARATION FOR DELIVERY

This section is not applicable to this specification.

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6. NOTES

6.1 Intended use. The conductive copper film covered by this specification is intended for use in coating electrically non-conductive materials for the purpose of subsequent electroplating.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification, MIS-15025, dated 10 March 1965.

Custodian:
Army-MI

Preparing Activity
Army-MI
Project No. 95GP-A009

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
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